

Measuring the gap between the Melnikov threshold and the period-doubling cascade in the damped driven pendulum

Elliot Jung

2026-06-11 · Pendulum Lab numerical laboratory (TypeScript engine, externally cross-validated)

Abstract. The damped driven pendulum $\theta'' = -\sin \theta - \gamma\theta' + A \cos(\omega t)$ ($\omega = 0.6667$) carries two distinct, frequently conflated notions of a “chaos threshold”: the analytic Melnikov amplitude $A_c(\gamma)$, above which first-order perturbation theory predicts a transverse homoclinic tangle, and the period-doubling onset $A_{PD}(\gamma)$, where the primary period-1 attractor begins the Feigenbaum cascade that produces the sustained chaotic attractor. We measure A_{PD} over $\gamma \in [0.1, 0.8]$ with an attractor-strobed bisection refined by the Floquet multiplier of the Newton periodic orbit (onset interpolated at $\rho = -1$), and compare it with the closed-form A_c . The ratio A_{PD}/A_c falls monotonically from 2.38 at $\gamma = 0.1$ to 0.987 at $\gamma = 0.8$, and the widely quoted ordering $A_c < A_{PD}$ *reverses* near $\gamma \approx 0.69$: at low damping the tangle precedes the cascade by a wide and rapidly growing margin, while at strong damping the cascade begins *below* the first-order Melnikov prediction. At the literature point $\gamma = 0.5$ our measurement $A_{PD} = 1.0664$ agrees with the published 1.0663 (Baker & Gollub) to four digits.

1. Introduction

The sinusoidally driven, damped pendulum is the canonical low-dimensional route to chaos, and it supports two different analytic/numerical landmarks as the drive amplitude A grows. The Melnikov method gives a closed-form first-order threshold $A_c = (4\gamma\omega_0/\pi) \cdot \cosh(\pi\omega/2\omega_0)$ above which the stable and unstable manifolds of the hilltop saddle intersect transversally, creating a Smale-horseshoe tangle. The tangle guarantees *transient* chaos and fractal basin boundaries — not a chaotic attractor. The sustained chaotic attractor instead appears at the end of a period-doubling cascade of the primary period-1 response, whose onset A_{PD} is a property of a specific attractor branch and has no closed form. Textbooks typically note $A_c < A_{PD}$ at the classic parameter point $\gamma = 0.5$, $\omega = 2/3$. This paper asks the quantitative question: *how does the gap between the two thresholds behave as damping is varied?*

2. Methods

All computations use the open Pendulum Lab engine, whose equations of motion are validated component-wise against an independent SymPy symbolic derivation (max relative deviation $\approx 10^{-14}$) and whose trajectories are cross-validated against SciPy DOP853 at $\text{rtol} = 10^{-13}$; the engine also reproduces published anchors (elliptic pendulum period, double-pendulum normal modes, and the $\gamma = 0.5$ period-doubling onset). The drive is made autonomous by carrying the phase as a third coordinate; integration is RK4 with $\text{dt} = 0.005$ snapped so an integer number of steps spans one drive period exactly.

A_{PD} measurement. For each γ , the drive amplitude is marched upward from $0.9 \cdot A_c$ with the strobed state warm-started at every step, so the measurement follows the physically realised attractor branch (this matters: the symmetric period-1 orbit pitchforks before the cascade, and a Newton continuation of the *symmetric* orbit would miss the doubling of the symmetry-broken branch the attractor actually follows). The loss of period-1 stability is bisected on the strobe map (transients of 300–600 drive periods; period detected in the $(\sin \theta, \cos \theta, \omega)$ embedding, immune to 2π winding). The bracket is then refined by Newton periodic orbits *seeded from the attractor*: the most negative real Floquet multiplier $\rho(A)$ of the orbit is interpolated through $\rho = -1$. A measurement is accepted as a period doubling only when this crossing is verified; otherwise the loss is reported as non-PD. Halving dt changes the $\gamma = 0.5$ onset by $|\Delta| \approx 7.1\text{e-}13$, so discretisation error is negligible at the quoted precision.

Corroboration. At $0.97 \cdot A_{PD}$ and $1.08 \cdot A_{PD}$ the Gottwald–Melbourne 0–1 test is applied to the $\cos \theta$ strobe series (700 samples): $K \approx 0$ confirms regular motion below onset; the value above onset depends on whether $1.08 \cdot A_{PD}$ lands beyond the cascade accumulation point or inside a periodic window, and is reported without prejudice.

3. Results

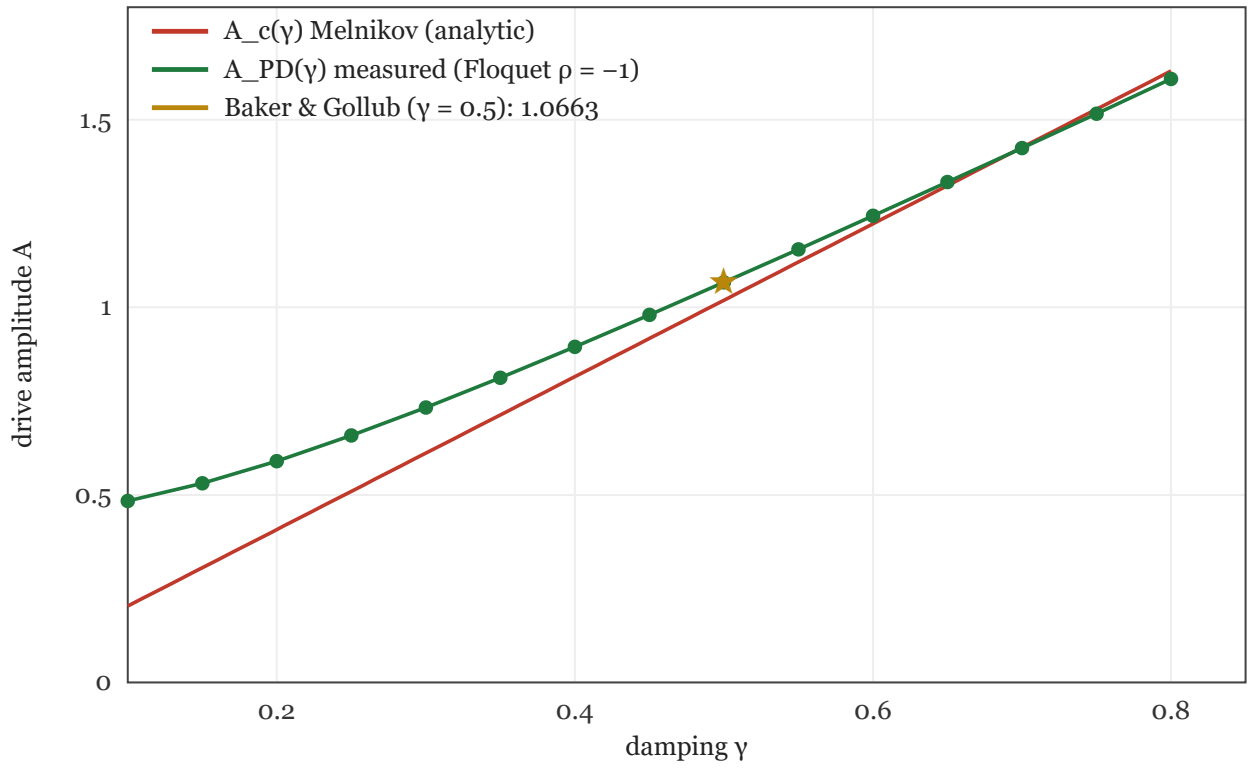


Figure 1. The analytic Melnikov threshold $A_c(\gamma)$ (red line) and the measured period-doubling onset $A_{PD}(\gamma)$ (green circles, Floquet-refined). The star is the published $\gamma = 0.5$ value 1.0663 (Baker & Gollub); our measurement at that point is 1.06637. A_c is exactly linear in γ ; the measured cascade onset is not.

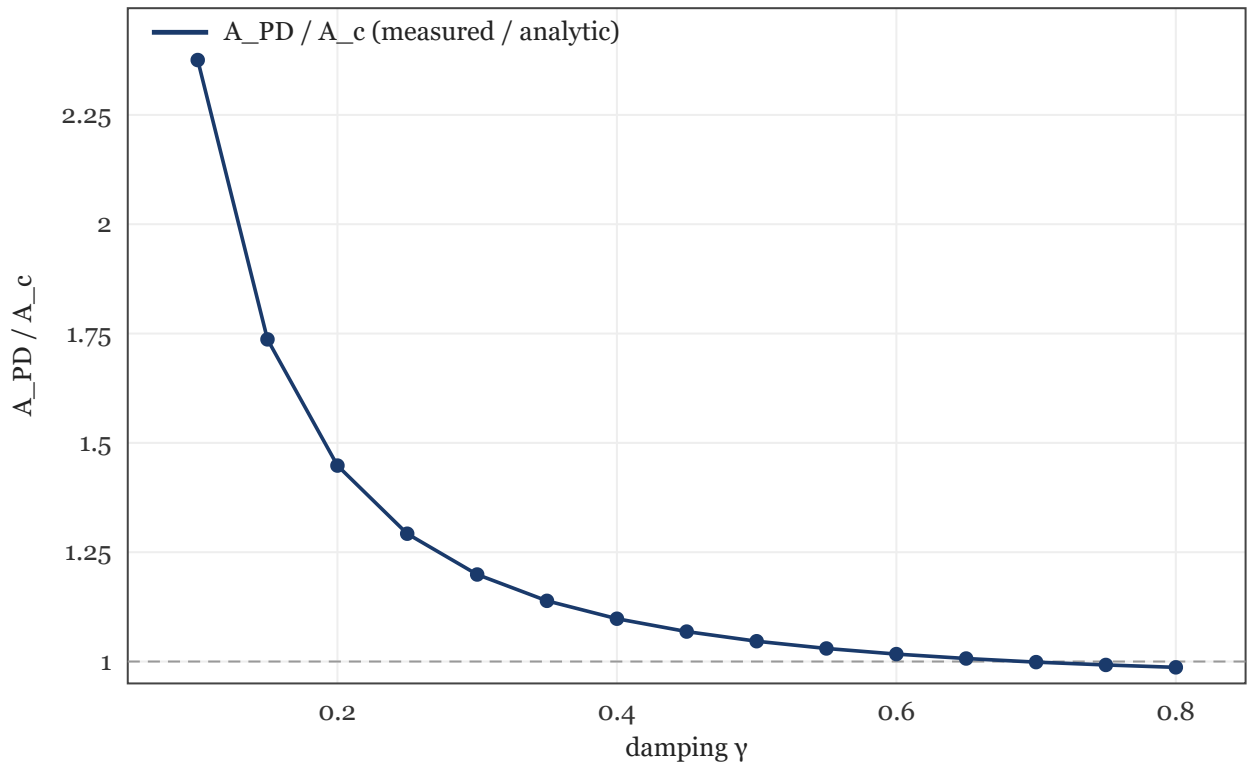


Figure 2. The ratio A_{PD}/A_c versus damping. The gap closes monotonically with increasing γ and crosses 1 near $\gamma \approx 0.69$: beyond this damping the period-doubling cascade of the primary attractor begins *below* the Melnikov tangle threshold. At $\gamma = 0.1$ the cascade requires 138% more drive than the tangle.

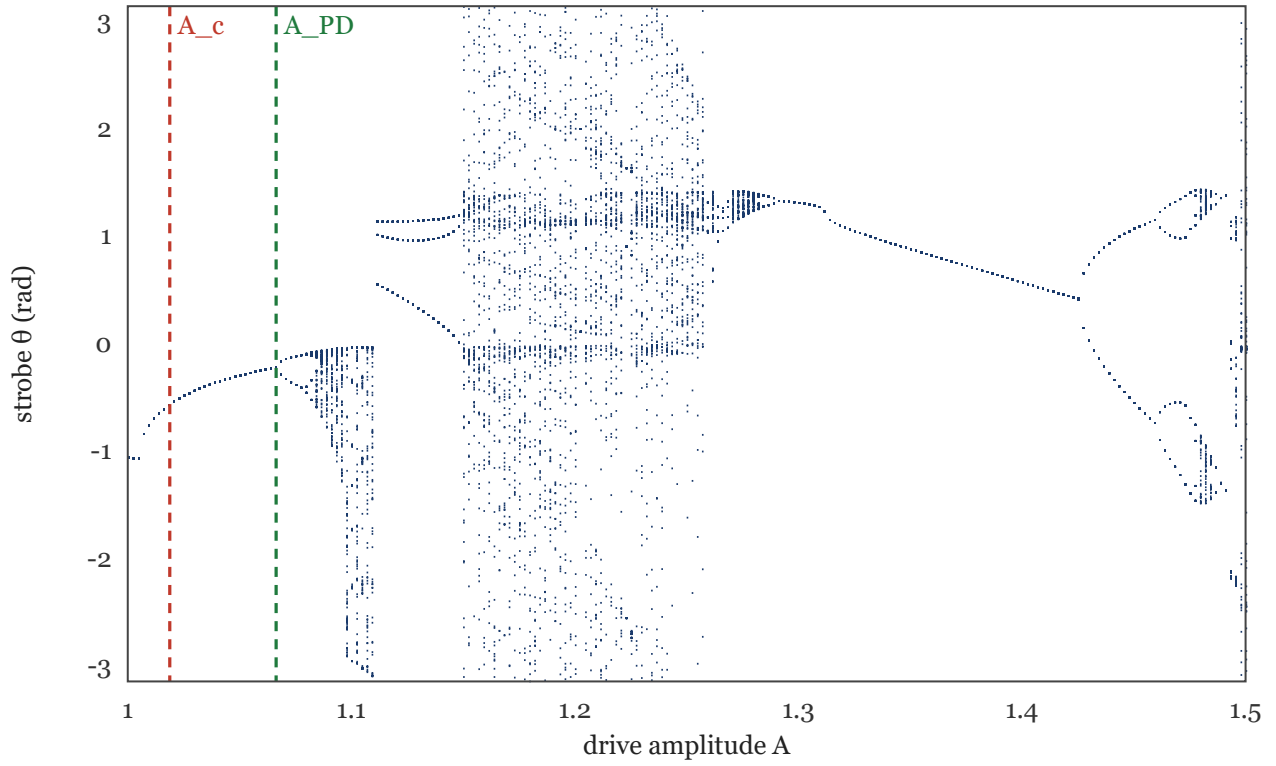


Figure 3. Strobe bifurcation diagram at $\gamma = 0.5$ (θ sampled once per drive period after a 250-period transient, warm-started in A). The dashed lines mark A_c (red) and the measured A_{PD} (green). Between them the motion is periodic but the phase space already contains the homoclinic tangle: transient chaos and fractal basin boundaries without a strange attractor. The cascade, chaotic band, and the large periodic (rotating) window are visible beyond A_{PD} .

γ	A_c (Melnikov)	A_{PD} (measured)	A_{PD}/A_c	loss of period-1	K at $0.97 \cdot A$	K at $1.08 \cdot A$
0.10	0.20375	0.48398	2.3753	PD ($\rho \rightarrow -1$)	0.02	1.00
0.15	0.30563	0.53080	1.7367	PD ($\rho \rightarrow -1$)	0.02	1.00
0.20	0.40751	0.59004	1.4479	PD ($\rho \rightarrow -1$)	0.02	-0.01
0.25	0.50939	0.65829	1.2923	PD ($\rho \rightarrow -1$)	-0.02	1.00
0.30	0.61126	0.73294	1.1991	PD ($\rho \rightarrow -1$)	0.02	0.06
0.35	0.71314	0.81217	1.1389	PD ($\rho \rightarrow -1$)	-0.02	1.00
0.40	0.81502	0.89470	1.0978	PD ($\rho \rightarrow -1$)	0.02	-0.01
0.45	0.91690	0.97964	1.0684	PD ($\rho \rightarrow -1$)	-0.02	-0.01
0.50	1.01877	1.06637	1.0467	PD ($\rho \rightarrow -1$)	-0.02	1.00

γ	A_c (Melnikov)	A_{PD} (measured)	A_{PD}/A_c	loss of period-1	K at $0.97 \cdot A$	K at $1.08 \cdot A$
0.55	1.12065	1.15448	1.0302	PD ($\rho \rightarrow -1$)	-0.02	1.00
0.60	1.22253	1.24368	1.0173	PD ($\rho \rightarrow -1$)	-0.01	1.00
0.65	1.32441	1.33377	1.0071	PD ($\rho \rightarrow -1$)	0.02	-0.01
0.70	1.42628	1.42463	0.9988	PD ($\rho \rightarrow -1$)	0.02	-0.57
0.75	1.52816	1.51622	0.9922	PD ($\rho \rightarrow -1$)	0.02	0.13
0.80	1.63004	1.60853	0.9868	PD ($\rho \rightarrow -1$)	0.02	1.00

4. Discussion

Three regimes emerge. (i) **Low damping ($\gamma \lesssim 0.2$):** $A_c \rightarrow 0$ linearly while the cascade onset of the primary resonance remains an order-one amplitude, so the ratio diverges (2.38 already at $\gamma = 0.1$). The window of “tangle but no strange attractor” — long chaotic transients and fractal basin boundaries below a still-periodic attractor — is widest here, and the phase space is visibly multistable (see §5). (ii) **Moderate damping:** the textbook ordering $A_c < A_{PD}$ holds, but the margin shrinks steadily (to 5% at $\gamma = 0.5$). (iii) **Strong damping ($\gamma \gtrsim 0.69$):** the measured cascade begins *below* the first-order Melnikov prediction. This is not a contradiction — the Melnikov threshold is asymptotically exact only as $\gamma, A \rightarrow 0$, and by $\gamma \approx 0.7$ the perturbation parameter is $O(1)$ — but it sharpens the usual caveat into a measured boundary: the first-order formula stops being even an ordering bound near $\gamma \approx 0.69$.

The 0–1 test values corroborate the structural picture: $K \approx 0$ on the period-1 side everywhere, while above onset K depends on where $1.08 \cdot A_{PD}$ falls relative to the cascade accumulation point and the periodic windows visible in Figure 3 — both outcomes occur in the table, as expected for a Feigenbaum scenario with embedded windows.

5. Limitations and reproducibility

The study fixes $\omega = 2/3$ and follows a single attractor branch per γ (warm-started in A); coexisting attractors reached from other initial conditions may double elsewhere. At the lowest dampings the phase space is multistable enough that a finer warm-started march can hop basins before the doubling — at $\gamma = 0.15$ a basin-capture transition of the followed state was observed near $A \approx 0.49$ (the orbit itself remains strongly stable there, $\rho \approx +0.29$),

below the verified doubling at 0.531. The quoted A_{PD} values are therefore specifically the $\rho = -1$ doublings of the primary oscillating branch, not necessarily the first event of any kind along a slow amplitude sweep. The Melnikov comparison concerns the first-order formula specifically — higher-order or numerical manifold computations would move A_c . The full study regenerates with `npm run paper:study` (~4 min) followed by `npm run paper:build`; the underlying engine, its 435+ unit tests, and the SymPy/SciPy cross-validations are in the same repository.

References

- G. L. Baker and J. P. Gollub, *Chaotic Dynamics: An Introduction*, 2nd ed., Cambridge University Press (1996).
- J. Guckenheimer and P. Holmes, *Nonlinear Oscillations, Dynamical Systems, and Bifurcations of Vector Fields*, Springer, §4.5 (1983).
- V. K. Melnikov, “On the stability of the center for time-periodic perturbations,” *Trans. Moscow Math. Soc.* **12**, 1–57 (1963).
- G. A. Gottwald and I. Melbourne, “On the implementation of the 0–1 test for chaos,” *SIAM J. Appl. Dyn. Syst.* **8**, 129–145 (2009).
- M. J. Feigenbaum, “Quantitative universality for a class of nonlinear transformations,” *J. Stat. Phys.* **19**, 25–52 (1978).